

CENG114 – Introduction to Programming II

Course Information

Lecturer: Yusuf Evren AYKAÇ

Office Hours: By appointment via email at evrenaykac@gmail.com

Teaching Assistants: Çağın ÖZKAYA, Hatice UYSAL, Yusuf Ekrem KEÇİLİOĞLU

Semester: Spring 2025/2026

Schedule: Thursdays, 11:30

Classroom: B413

Credits: 6 (3 hours of theoretical lectures + 2 hours of laboratory sessions per week)

Course Description

This course is a continuation of CENG113 (Introduction to Programming 1) and covers sorting and searching algorithms, recursion, and object-oriented programming (OOP) in Java, focusing on fundamental principles such as encapsulation, inheritance, polymorphism, abstract classes, interfaces, exception handling, and text/binary I/O. The course further explores generics, data structures (lists, stacks, queues, linked lists), and introduces event-driven programming. By the end of the course, students will be able to design, implement, test, and document medium-scale software applications following object-oriented design principles.

Textbook & References

Primary Textbook:

- Liang, Y. Daniel. Introduction to Java Programming and Data Structures, Comprehensive Version, 13th Edition, Pearson.

Additional References:

- Cay S. Horstmann, Core Java Volume I – Fundamentals, 12th Edition, Pearson, 2022.
- Joshua Bloch, Effective Java, 3rd Edition, Addison-Wesley, 2018.

Grading Policy

Component	Weight
Midterm Exam (1)	20%
Final Exam (1)	25%
Lab Quizzes (3)	30%
Lab Performance Grade (1)	10%
Lab Exam (1)	15%

Letter grades will be assigned based on the official AYBU catalogue grading system.

Attendance and Make-up Policy

Attendance will be taken regularly and must comply with AYBU regulations on absenteeism. Make-up exams will be provided **only** for the midterm and final exams upon submission of an official excuse. No make-ups will be given for quizzes or lab exams.

Tentative Weekly Schedule

	Lecture Topics	Lab
Week 1: 12.02.2026	Sorting and Searching Algorithms	No Lab
Week 2: 19.02.2026	Recursion: versus iteration, exercises (factorial, fibonacci, etc.)	Lab 1
Week 3: 26.02.2026	Objects and Classes	Lab 2
Week 4: 05.03.2026	Object-Oriented Thinking	Lab Quiz – 1
Week 5: 12.03.2026	Inheritance / Polymorphism & OOD	Lab 3
Week 6: 19.03.2026	Holiday	Lab 4
Week 7: 26.03.2026	Abstract Classes and Interfaces	Lab 5
Week 8: 02.04.2026	Design Principles & Metrics	Lab Quiz – 2
Week 9: 09.04.2026	Midterm Exam	
Week 10: 16.04.2026	Exception Handling & Text I/O & Event-Driven Programming	Lab 6
Week 11: 23.04.2026	Holiday	Lab 7
Week 12: 30.04.2026	Generics and Collections	Lab Quiz – 3
Week 13: 07.05.2026	Linked Lists	Lab 8
Week 14: 14.05.2026	Linked Lists / Stacks and Queues	Lab 9
Week 15: 21.05.2026	Lambda Expressions & Streams / Review	Lab Exam